BTechi

IIT Madras BS

PYQ PATTERN · STATS I · SEP 2023

Most students pick the wrong answer here

One line. One trap. Answer on slide 3.

Swipe →

STATS I · MEAN

Your turn

The average annual package is ₹8 lakh. From each value, 5 is subtracted, and then the result is doubled. What is the new average?

A ₹11 lakh

B ₹6 lakh

C ₹3 lakh

D ₹16 lakh

Comment your answer, then swipe.

ANSWER

B. ₹6 lakh

1. Subtract 5 from every value: the mean also drops by 5. 8 – 5 = 3.
2. Double every value: the mean also doubles. 2 × 3 = 6.

⚠ Order matters: 2(x̄ – 5) is not 2x̄ – 5. The wrong order gives ₹11 lakh.

BONUS

What happens to the spread?

- Adding or subtracting a number never changes the SD, variance, range or IQR.
- Multiplying by 2 doubles the SD and multiplies the variance by 4.

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